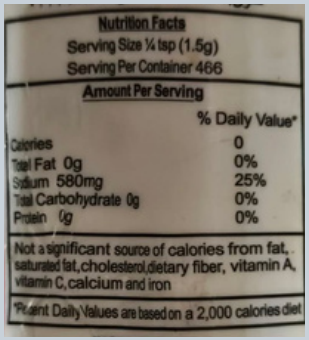
Atom count card: Sea Salt (NaCl)



A: Product identity

Product: Sea salt, Riyadh Salt Industry Co.
Compound: sodium chloride, NaCl

Why it's here: NaCl is the seasoning itself, and the label reports its sodium

Student: Kasey Carlyle D. Guadalupe, Chemistry for Engineers

B: Base quantitative data

Quantity	Value	Source	Precision
Sodium per serving	580 mg	Product label	2 s.f.
Serving size	1.5 g (1/4 tsp)	Product label	2 s.f.
Servings per container	466	Product label	Exact
Formula	NaCl	PubChem	Exact
Atomic masses	Na 22.990 g/mol Cl 35.453 g/mol	Science Notes periodic table (2017)	Not limiting
Avogadro constant	$6.022\,140\,76 \times 10^{23}$ mol^{-1}	BIPM (2019)	Exact

C: Molar-mass build-up

Element	Atoms	g/mol each	g/mol	% by mass
Na	1	22.990	22.990	39.337
Cl	1	35.453	35.453	60.663
NaCl	2		58.443	100.000

Both contributions have 3 decimal places, so M keeps 3 decimal places. Percentages sum to 100.000%.

D: The mole map

Grams never convert straight to particles. Always pass through moles, and name the entity at every step.

6.9×10^{2} g NaCl —÷58.443 g/mol→ 1.2×10^{1} mol NaCl

1.2×10^{1} mol NaCl —× 6.022 140 76 × 10^{23} formula units/mol→ 7.1×10^{24} NaCl formula units

7.1×10^{24} NaCl formula units —×1 Na atom per formula unit→ 7.1×10^{24} Na atoms

E: Mole-conversion calculations

1. Molar mass of NaCl
1(22.990) + 1(35.453) = 58.443 g/mol

2. Label bridge: sodium → NaCl in the container
580 ~~mg Na~~ × (1 g/1000 ~~mg~~) × (58.443 g NaCl/22.990 ~~g Na~~) = 1.4744 g NaCl per serving
1.4744 ~~g/serving~~ × 466 ~~servings~~ = 687.08 g → 6.9×10^{2} g NaCl

3. Mass → moles
687.08 ~~g NaCl~~ × (1 mol NaCl / 58.443 ~~g NaCl~~) = 11.7564 mol → 1.2×10^{1} mol NaCl

4. Moles → formula units
11.7564 ~~mol NaCl~~ × (6.022 140 76 × 10^{23} formula units / 1 ~~mol~~) = 7.0799×10^{24} → 7.1×10^{24} NaCl formula units

NaCl is ionic, so I count formula units, not molecules

5. Atoms of one element
7.0799×10^{24} ~~formula units NaCl~~ × (1 Na atom / 1 ~~formula unit~~) = 7.0799×10^{24} → 7.1×10^{24} Na atoms

6. Percent composition and two-route check (Na)
Route A: %Na = (22.990 / 58.443) × 100% = 39.337% 687.08 g × 0.39337 = 270.3 g Na
Route B: 11.7564 mol NaCl × (1 mol Na / 1 mol NaCl) × (22.990 g Na / 1 mol Na) = 270.28 g Na
Both routes → 2.7×10^{2} g Na ✔ agree

Both routes give 2.7×10^{2} g of sodium, so my mole steps are consistent.

7. Reverse: % composition → formula
In 100 g: Na 39.337 g ÷ 22.990 = 1.711 mol; Cl 60.663 g ÷ 35.453 = 1.711 mol
÷ 1.711 → Na 1 : Cl 1 → NaCl (empirical = molecular, n = 58.443 ÷ 58.443 = 1)

8. Scale-up and reverse
40 ~~containers~~ × (11.7564 mol / 1 ~~container~~) = 470.26 mol → 4.7×10^{2} mol NaCl
1.00 ~~mol~~ × (58.443 g / 1 ~~mol~~) × (1 serving / 1.4744 g) = 39.638 → 4.0×10^{1} servings per mole

One mole is only 4.0×10^{1} servings, so one mole is a small percentage of one container.

F: Scale-up, from one container to a class

Assumed scenario: 40 students × 1 container each. The count is exact.

40 containers is about 27kg of salt.

One container:
- 1.2×10^{1} mol NaCl
- 7.1×10^{24} NaCl formula units
- 6.9×10^{2} g NaCl

40 containers
- 4.7×10^{2} mol NaCl
- 2.7×10^{4} g (about 27 kg)
- 2.8×10^{26} NaCl formula units

Reverse, how much makes one mole?
- 4.0×10^{1} servings make 1.00 mol
- That's 8.5×10^{-2} of one container (1.00 mol ÷ 11.7564 mol)

G: Sig-fig check

Calculator: 7.0799×10^{24} formula units
Reported: 7.1×10^{24} NaCl formula units
What limits it: The label's sodium value, 580 mg, taken as 2 s.f. (5.8×10^{2} mg)
Exact, so not limiting: N_a, formula subscripts, 1 g = 1000 mg, 466 servings, 40 containers
Not limiting: molar masses and percentages from atomic masses (carried at 4 to 5 s.f.) Guard digits kept until the last line of each calculation.

Takeaway

One container of salt contains about 7.1×10^{24} NaCl formula units, which surprised me because for less than a kilogram of salt, it's an incredibly large number of tiny particles.

AI Disclosure

AI (Claude) was used to explain mole-concept steps, check my calculations as a second opinion, and give feedback on my card layout and the wording of my takeaway sentence. All label values, calculations, references and final answers were independently verified by me.

Bureau International des Poids et Mesures. (2019). *The International System of Units (SI)* (9th ed.). https://www.bipm.org/en/publications/si-brochure

National Center for Biotechnology Information. (n.d.). *PubChem compound summary for CID 5234, sodium chloride*. Retrieved October 15, 2026, from https://pubchem.ncbi.nlm.nih.gov/compound/5234

Riyadh Salt Industry Co. (2025). *Risa Iodized Salt, 700 g* [Product label]. Photograph by Kasey Carlyle D. Guadalupe, October 13, 2026.

Science Notes. (2017). *Periodic table of the elements* [Image]. Retrieved October 13, 2026, from https://sciencenotes.org/wp-content/uploads/2017/01/PeriodicTableBW2017.png